

Morusin induces apoptosis and autophagy via JNK, ERK and PI3K/Akt signaling in human lung carcinoma cells

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ABSTRACT

Due to drug resistance and side effects, the development of novel therapeutics for the treatment of lung cancer is still in an urgent need. Morusin, a naturally occurring prenylated flavonoid isolated from the root bark of *Morus alba*, has been reported to be a promising candidate for cancer treatment including lung cancer. This study aimed to validate the anti-cancer effects of morusin in human non-small cell lung cancer (NSCLC) cell lines A549 and NCI-H292. The results indicated that morusin had growth inhibitory, pro-apoptotic and pro-autophagic effects on A549 and NCI-H292 cells. The induction of apoptosis was characterized by chromatin condensation and PARP cleavage. Mitochondrial membrane potential (MMP) loss, cytochrome *c* release, Bax/Bcl-2 dysregulation, and caspase-3 cleavage were also observed, indicating a mitochondria-dependent apoptosis was induced by morusin. A pro-autophagic effect was demonstrated by the increased level of LC3-II and decreased level of SQSTM1/p62. Furthermore, morusin inhibited PI3K/Akt signaling and activated JNK, ERK pathways as indicated by the alteration in the ratio of phosphorylation level over total protein expression level. A PI3K/Akt inhibitor (LY294002), a JNK inhibitor (SP600125) and a MEK/ERK inhibitor (U0126) contributed to the determination that these pathways were involved in both apoptosis and autophagy induced by morusin. Moreover, morusin treatment strikingly enhanced intracellular ROS level, an ROS scavenger NAC blocked cell death and changes of Akt, JNK and ERK induced by morusin.

1. Introduction

Cancer has been a major public health problem worldwide for many years. Lung cancer is considered as the most common type of cancer and by far the leading cause of cancer-related deaths in the world. According to cancer statistics released by the American Cancer Society, approximately 228,820 new lung cancer cases and 135,720 lung cancer deaths were projected to occur in the United States in 2020. The 5-year relative survival rate for lung cancer is only 19%, which is among the lowest of all cancer types [1]. Treatment options for lung cancer include surgery, chemotherapy, radiation therapy, immunotherapy and targeted therapy. Despite the improvements in therapeutics especially novel molecular-targeted approaches emerged during the past decades, the death rate of lung cancer remains the highest among all cancer types [2]. Therefore, development of novel therapies is still in an urgent need.

Natural products have been demonstrated as the most prolific source of lead compounds for the development of anti-cancer drugs [3].

Prenylated flavonoids represented a subgroup of natural occurring flavonoids possessing at least one C5 isoprene unit in their structure. The introduction of lipophilic prenyl side chain(s) into flavonoid skeleton might enhance interactions between flavonoids and the phospholipid membranes or hydrophobic target proteins, consequently increase the bioactivities of flavonoids [4]. Increasing evidences suggested that prenylation was a determinant factor for the enhancement of cytotoxicity of flavonoid, this made prenylated flavonoids as attractive compounds for obtaining anti-cancer leading structures [5]. The root bark of *Morus alba* (mulberry tree), named as *Sangbaipi* in Chinese, was demonstrated to be a good resource of prenylated flavonoids. During our exploration of cytotoxic prenylated flavonoids, we performed a bioassay guided isolation of *Sangbaipi*, morusin was obtained together with many other compounds [6]. Among the obtained compounds, morusin was the most effective and abundant one. It has been reported that morusin could induce the apoptosis of colorectal [7], lung [8], breast [9], prostate [10], gastric [11] and hepatoma [12] cancer cells. The mechanism

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of action involved in suppression of NF- κ B [7,12], STAT3 [12], EGFR/STAT3 [8] and c-myc [11]. Morusin also has been reported that it could induce paraptosis-like cell death in epithelial ovarian cancer cells [13]. In this study, the effects of morusin on cell growth in two human lung cancer cell lines, A549 and NCI-H292, were explored. The results indicated that morusin inhibited lung cancer cell growth in dose- and time-dependent manners. The effects appeared to occur through the PI3K/Akt, JNK and ERK signaling involved apoptosis and autophagy. JNK and ERK are members of MAPK signaling pathway. MAPK and PI3K/Akt pathways are two important intracellular pathways that transmit extracellular signals into cells [14,15]. These two pathways also regulate or have crosstalk with other important signaling such as NF- κ B and STAT3 [16], through these intracellular signaling, cell survival and death are regulated [17]. In this study, the regulation of MAPK and PI3K/Akt pathways by morusin in A549 and NCI-H292 cells were focused. Moreover, a strikingly enhanced intracellular ROS level induced by morusin played an important role in both apoptosis and autophagy.

2. Materials and methods

2.1. Chemicals and reagents

Morusin was purified by our laboratory from the root bark of *M. alba* (Sangbaipi) as described previously (Fig. 1A) [6], the purity was determined to be 97% by HPLC analysis. Primary antibodies against Bax (Cat# 60267-1-Ig), Bcl-2 (Cat# 12789-1-AP), Bcl-x_L (Cat# 10783-1-AP), caspase-3 (Cat# 19677-1-AP), SQSTM1 (Cat# 18420-1-AP), cytochrome c (Cat# 10993-1-AP), and LC3 (Cat# 14600-1-AP) were purchased from Proteintech Group. Primary antibodies against cleaved caspase-3 (Cat# 9669), cleaved caspase-9 (Cat# 9505) PARP (Cat# 9542), NF- κ B p65 (Cat# 8242) and Phospho-NF- κ B p65 (Cat# 3033) were purchased from Cell Signaling Technology. Primary antibodies against GAPDH (Cat# sc-25778), β -actin (Cat# sc-47778), ERK (Cat# sc-514302), p-ERK (Cat# sc-7383), p-JNK (Cat# sc-6254), JNK (Cat# sc-7345), p-Akt1/2/3 (Cat# sc-271964) and Akt (Cat# sc-81434) were purchased from Santa Cruz Biotechnology. Anti-rabbit IgG (Cat# SA00001-2) and anti-mouse IgG (Cat# SA00001-1) were from Proteintech Group. U0126 and LY294002 were purchased from Selleckchem. SP600125, Z-VAD-FMK, N-acetyl-cysteine (NAC) and Cell Counting Kit-8 (CCK-8) were

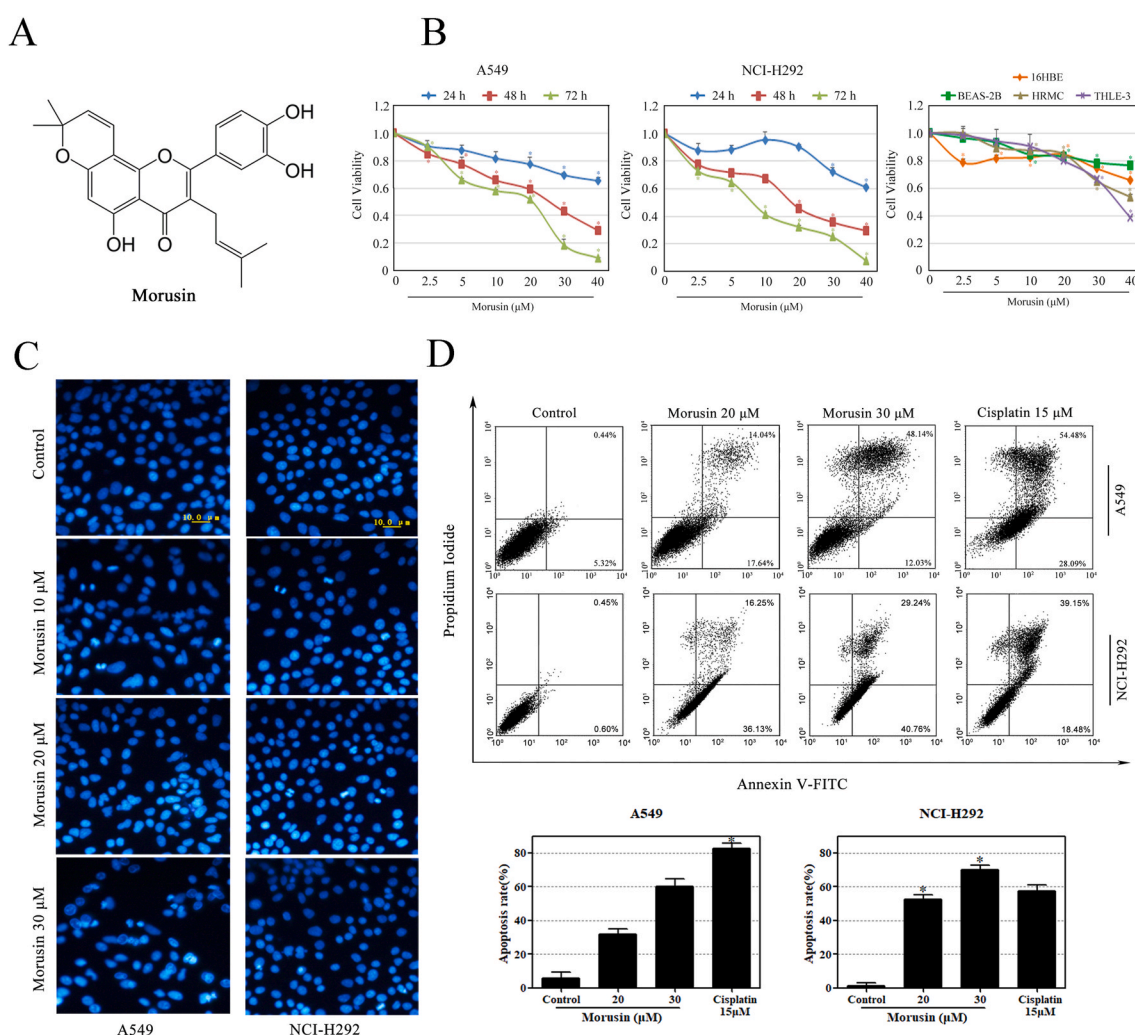


Fig. 1. Morusin inhibited cell proliferation and induced apoptosis in A549 and NCI-H292 cells. (A) Chemical structure of morusin. (B) A549, NCI-H292, 16HBE, BEAS-2B, HRMC and THLE-3 cells were treated with morusin as indicated and the inhibition of proliferation was detected by MTT assay. The 16HBE, BEAS-2B, HRMC and THLE-3 cells were treated with morusin for 48 h. Data are normalized to the control group. * $p < 0.05$ compared with control group. (C) A549 and NCI-H292 cells were treated with morusin for 24 h, the changes in nuclear morphology were evaluated by fluorescence microscopy with DAPI staining. (D) Cells were treated with morusin at indicated concentration for 24 h, stained with Annexin V/PI and followed by flow cytometry assay. Cisplatin 15 μ M treatment was used as positive control. Data are expressed as mean $\pm$ SD, $n = 3$. * $p < 0.05$ compared with control group.

purchased from MedChem Express. 3-(4, 5-dimethyl-2-thiazol)-2, 5-diphenyl-2-H-tetrazolium bromide (MTT) and chloroquine were from Solarbio. Annexin V-FITC/Propidium iodide (PI) apoptosis detection kit was from BD Biosciences. 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) and 4,6-diamidino-2-phenylindole (DAPI) were purchased from Sigma-Aldrich. 5,5',6,6'-tetrachloro-1,1',3,3'-tetraethyl benzimidazolyl-carbocyanine iodide (JC-1) was purchased from Cayman Chemical. Carbonyl cyanide m-chlorophenylhydrazone (CCCP) was from Alfa Aesar. Tandem mRFP-GFP-LC3 fluorescent plasmid was obtained from Addgene.

2.2. Cell culture

Human lung cancer A549 and NCI-H292 cells were purchased from the Shanghai Cell Bank and were cultured in Roswell Park Memorial Institute (RPMI) 1640 (Gibco, Grand Island, NY, USA) supplemented with 10% (v/v) fetal bovine serum (FBS, HyClone, Logan, UT, USA), 100 U/ml penicillin and 100 U/ml streptomycin. Human bronchial epithelial 16HBE and BEAS-2B cells were purchased from the Shanghai Cell Bank; the human renal mesangial cell line HRMC and immortalized human liver cell line THLE-3 were purchased from ATCC. They all were cultured in Dulbecco's Modified Eagle Medium (DMEM) (Gibco, Grand Island, NY, USA) containing 10% FBS. All cells were maintained at 37 °C in a humidified incubator containing 5% CO₂.

2.3. Cell viability assay

Cells were seeded at a density of 1×10^4 cells per well in 96-well plates and incubated overnight. The cells were then treated with DMSO or morusin for different times, cell viability was detected by MTT or CCK-8 assay. For MTT assay, after treatment, 20 µl of 2.0 mg/ml MTT solution was added in each well and incubated at 37 °C for an additional 4 h. The plates were centrifuged and the medium was removed, then 100 µl of DMSO was added to each well to dissolve the formazan crystals. The optical density value was measured at 570 nm with a microplate reader (Biorad, Model 680, USA). For CCK-8 assay, after treatment, 10 µl of CCK-8 solution was added in each well and incubated for an additional 4 h at 37 °C. The optical density value was measured at 450 nm. Triplicate wells were used for each sample and the experiments were repeated at least three times to get means and standard deviations [18].

2.4. Apoptosis assay

Apoptosis was detected by flow cytometry using an Annexin V-FITC/PI kit according to the manufacturer's instruction. In brief, after treatment with morusin for 24 h, cells were harvested, washed, centrifuged and then resuspended in binding buffer containing annexin V-FITC and PI. After staining, cells were analyzed using a FACS Calibur flow cytometer (BD Biosciences, San Jose, CA, USA). Cells treated with 15 µM cisplatin were served as positive control.

2.5. Autophagic flux measurement

A549 and NCI-H292 cells cultured on coverslips were transiently transfected with tandem mRFP-GFP-LC3 plasmid using lipofectamine 2000 accordingly to the manufacture's protocol. After the indicated treatments, cells on coverslips were fixed, washed, and then were visualized by fluorescence microscope. Representative cells were selected and photographed.

2.6. Intracellular ROS measurement

The intracellular ROS level was determined using DCFH-DA as fluorescence probe. After being treated with morusin for 24 h, cells were washed twice with PBS and loaded with 10 µM DCFH-DA for 30 min at

37 °C, then the cells were collected and analyzed by a flow cytometer at an excitation wavelength of 488 nm and an emission wavelength of 530 nm. Cells treated with H₂O₂ were used as positive control.

2.7. DAPI (4, 6-diamidino-2-phenylindole) staining

DNA condensation was detected with DAPI staining. Briefly, cells cultured in 24-wells plate were treated with morusin for 24 h, after fixation, cells were incubated with 2.5 µg/ml DAPI solution for 10 min in the dark. After washing with PBS, images of cell nuclei were taken by a fluorescence microscope system (Olympus, IX71, Japan).

2.8. Mitochondrial membrane potential (MMP) measurement

MMP was analyzed by 5, 5', 6, 6'-tetrachloro-1, 1', 3, 3'-tetraethyl benzimidazolyl-carbocyanine iodide (JC-1) staining. Cells cultured in 6-well plates were treated with different concentrations of morusin for 24 h, after washing with PBS, cells were then incubated with 2 µM JC-1 for 30 min. Cells were harvested, washed, centrifuged and then resuspended in PBS and analyzed using a FACS Calibur flow cytometer. Cells treated with 15 µM CCCP were served as positive control.

2.9. Immunoblot analysis

After treatments, cell lysate was prepared by using a sample buffer (50 mM Tris-HCl [pH 6.8], 2% SDS, 10% glycerol, 100 mM DTT, 0.1% bromophenol blue). Equal amounts of proteins were loaded and electrophoresed on SDS PAGE gel followed by a transfer onto nitrocellulose membrane (Millipore). After blocking, the membrane was incubated with a diluted primary antibody solution at 4 °C overnight, and then the membrane was exposed to a horseradish peroxidase-conjugated secondary antibody at room temperature for 1 h. Signals were visualized using the enhanced chemiluminescence (ECL) detection reagents (Millipore, Billerica, MA, USA). β-actin and GAPDH were used as internal reference [19].

2.10. Statistical analysis

Results are presented as the mean ± standard deviation (SD). To determine the significant difference between two groups, one way analysis of variance (ANOVA) test and post hoc multiple comparison Bonferroni test were used. $p < 0.05$ was considered to be significant [20].

3. Results

3.1. Effect of morusin on A549 and NCI-H292 cell viability

To determine the cytotoxicity of morusin on lung cancer cells, A549 and NCI-H292 cells were treated with different concentrations of morusin (0, 2.5, 5, 10, 20, 30, 40 µM) for different time periods (24, 48 and 72 h). As shown in Fig. 1B, increasing concentrations of morusin for different time intervals up to 72 h inhibited cell growth in a dose- and time-dependent manner in both A549 and NCI-H292 cells. The IC₅₀ values of morusin treatment for 48 h were 12.32 µM and 7.92 µM on A549 and NCI-H292 cells respectively. Meanwhile, two kinds of cells which derived from normal human bronchial epithelium, 16HBE and BEAS-2B, a human renal mesangial cell line HRMC and an immortalized human liver cell line THLE-3 were used for evaluating the selective cytotoxicity of morusin. In all the four kinds of normal cell lines, the cell growth after 48 h of morusin treatment was only slightly inhibited (Fig. 1B), indicating a selective effect of morusin against cancer cells.

3.2. Morusin induced caspase-dependent apoptosis in A549 and NCI-H292 cells

Apoptosis is a common type of cell death. In order to investigate whether cell apoptosis was involved in morusin induced cell viability reduction, DAPI staining and Annexin V-FITC/PI double staining were performed. As shown in Fig. 1C, morusin treated cell groups had more condensed and fragmented nuclei than untreated cell group. Moreover, the flow cytometric evaluation results indicated that the proportion of Annexin V-FITC positive cells increased dose-dependently after 24 h administration of morusin in both A549 and NCI-H292 cells (Fig. 1D). These results suggested that morusin induced cell death in A549 and NCI-H292 cells were involved in apoptotic events.

Furthermore, the expression levels of apoptosis-related proteins were

examined by immunoblot analysis. As shown in Fig. 2A, after morusin treatment, the cleavage of PARP, caspase-3 and caspase-9 increased dose-dependently, demonstrating that morusin induced apoptosis might be correlated with caspase cascade in both A549 and NCI-H292 cells. A pan-caspase inhibitor Z-VAD-FMK was employed to assess the function of caspase in morusin induced apoptosis. As shown in Fig. 4D, co-treatment of morusin with Z-VAD-FMK reduced the expression of cleaved PARP, caspase-3 and caspase-9. Moreover, the reduction of cell viability induced by morusin was partially reversed by co-treatment with Z-VAD-FMK (Fig. 4C). These results confirmed that morusin induced caspase-dependent apoptosis in A549 and NCI-H292 cells.

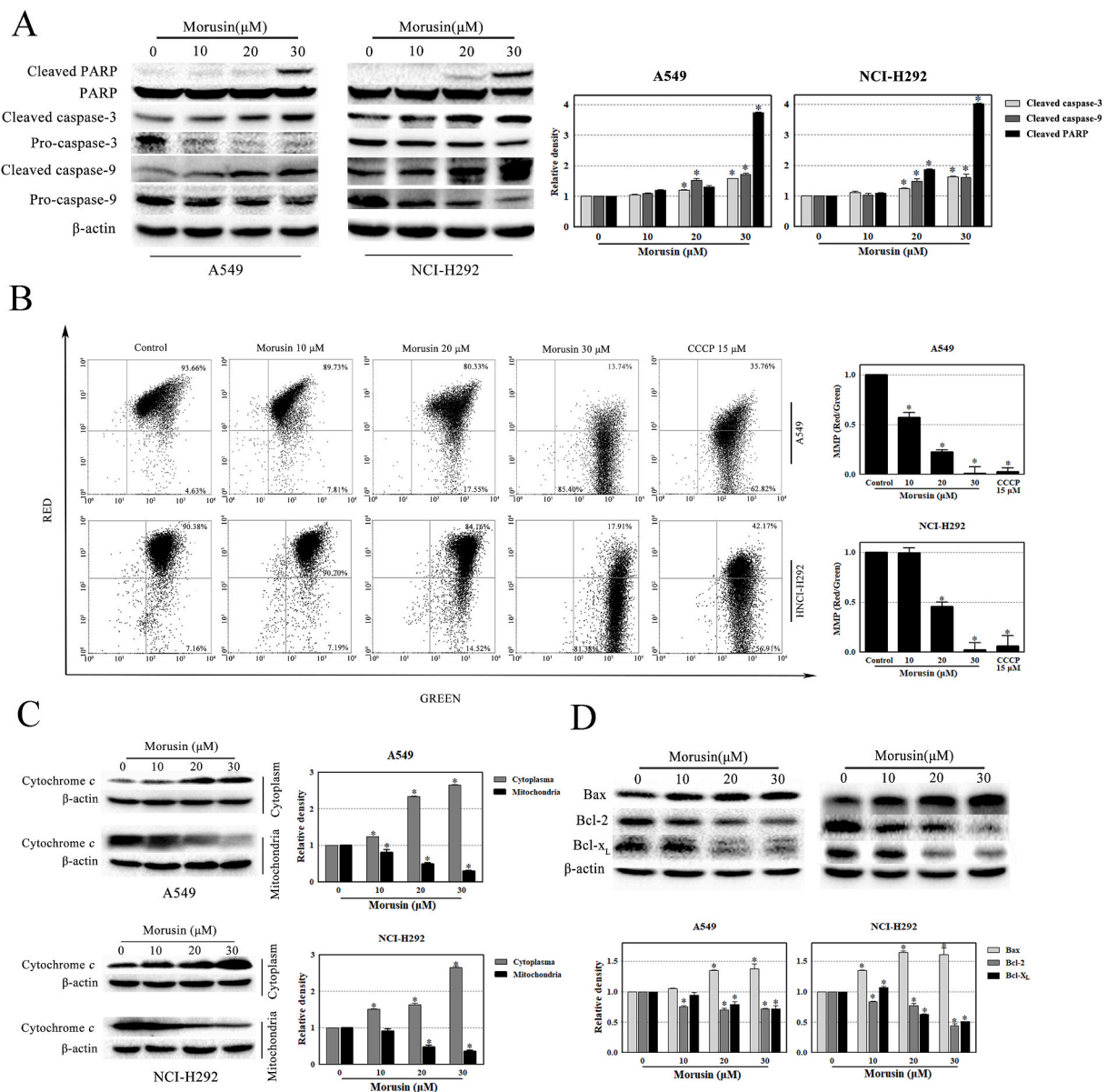


Fig. 2. Morusin induced endogenous apoptosis in A549 and NCI-H292 cells. Cells were incubated with morusin for 24 h. (A) Cell lysates were subjected to western blot for caspase-3, caspase-9 and PARP, β-actin was used as loading control. The band density was normalized to untreated group. * $p < 0.05$ compared with control group. (B) MMP changes were monitored by loading with JC-1 and detected by flow cytometry. CCCP 15 μM treatment was used as positive control. Cell populations indicating JC-1 aggregates (red) are in the upper right quadrant and JC-1 monomers (green) are in the lower right part. Bar graph represented the mean $\pm$ SD, $n = 3$. * $p < 0.05$ compared with control group. (C) The protein levels of cytochrome c in both mitochondria and cytoplasm were determined by immunoblot analysis. (D) Bax, Bcl-2 and Bcl-xL were detected by immunoblotting. β-actin was used as loading control. The band density was normalized to untreated group. * $p < 0.05$ compared with control group.

3.3. Morusin induced apoptosis through mitochondrial pathway in A549 and NCI-H292 cells

The caspase cascades may be triggered by mitochondrial membrane depolarization and release of mitochondrial proteins such as cytochrome *c* [21]. The changes of MMP in A549 and NCI-H292 cells were detected first. After treatment with morusin for 24 h, JC-1 staining was used for the analyzing of MMP. Flow cytometric analysis revealed that there was a high ratio of red to green fluorescence intensity in control group, while the ratios decreased dramatically in the morusin treated groups. The positive control, CCCP treated cells, showed a very small red to green ratio (Fig. 2B). These results indicated that morusin treatment induced mitochondrial depolarization and loss of MMP in the two lung cancer cell lines.

Cytochrome *c* is normally located in mitochondrial intermembrane space, when mitochondrial depolarization and loss of MMP occurred, cytochrome *c* will be released from mitochondria into cytoplasm [22]. The cytochrome *c* protein level in the subcellular compartments was

then detected by immunoblot analysis. As shown in Fig. 2C, there was a significant upregulation of cytochrome *c* in the cytosolic fraction and downregulation in the mitochondria fraction after morusin exposure for 24 h, which indicated a morusin-induced cytochrome *c* translocation occurred in both A549 and NCI-H292 cells.

Cytochrome *c* translocation usually indicates an early event leading to apoptosis [23]. It has been reported that the mitochondrial pathway of apoptosis is dependent on the Bcl-2 protein family, which control apoptosis by governing mitochondria outer membrane permeabilization [24]. We analyzed the expression of Bcl-2 family proteins in A549 and NCI-H292 cells treated with morusin by immunoblot. As a result, the expression of pro-apoptotic protein, Bax, increased dose-dependently; meanwhile, the levels of anti-apoptotic proteins Bcl-2 and Bcl-X_L decreased (Fig. 2D).

3.4. Morusin induced autophagy in A549 and NCI-H292 cells

Autophagy is another mode of programmed cell death and usually

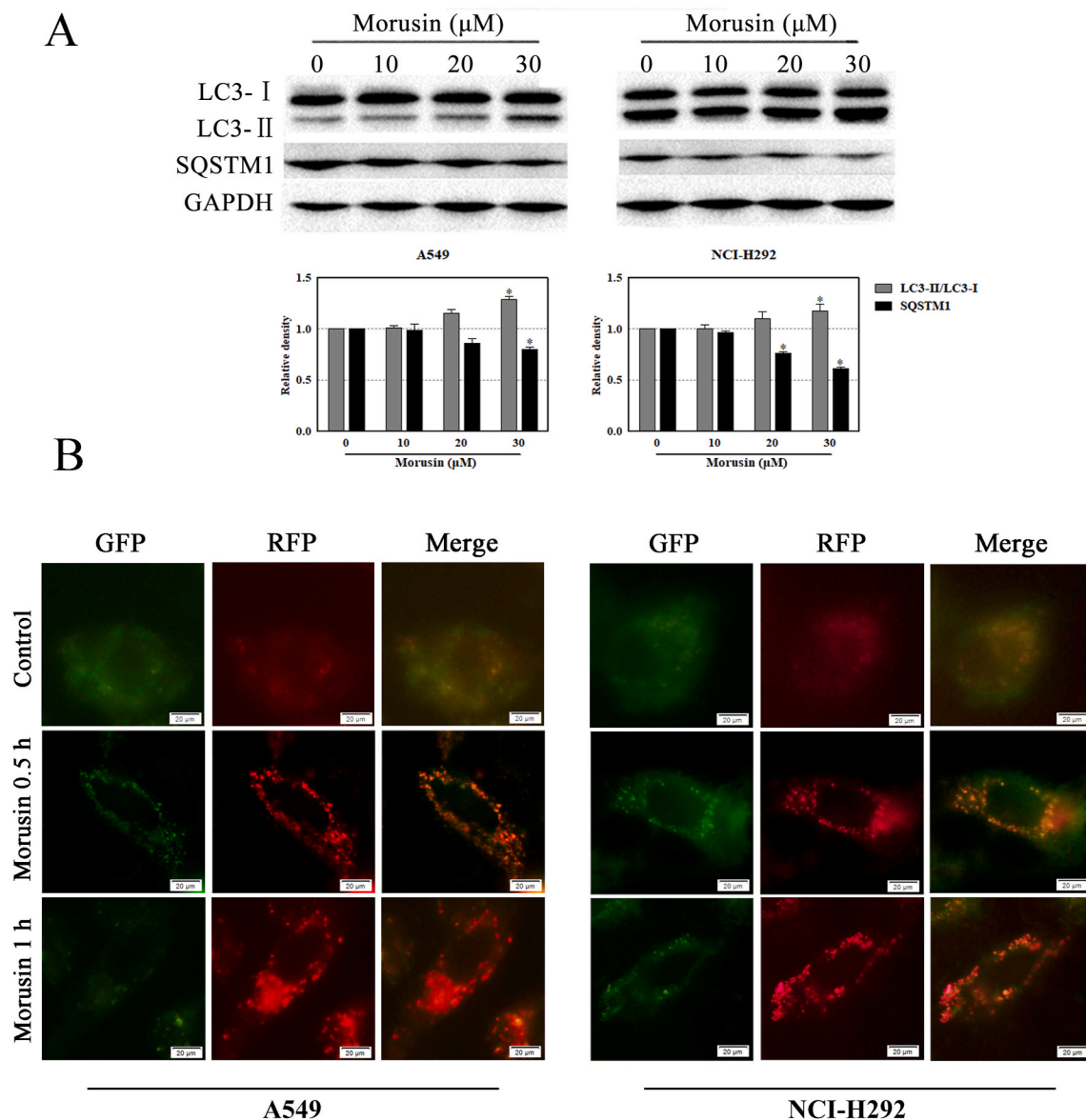


Fig. 3. Morusin induced autophagy in A549 and NCI-H292 cells. (A) Cells were incubated with morusin at indicated doses for 0.5 h, cell lysates were subjected to immunoblotting for LC3 and SQSTM1, and GAPDH was used as control. Bar graphs represented the relative express level of proteins. Values are expressed as mean $\pm$ SD, $n = 3$, * $p < 0.05$ versus control group. (B) Cells transfected with tandem RFP-GFP-LC3 plasmid were incubated with morusin at 20 μ M for indicated times, then observed under fluorescence microscopy.

acts ahead of apoptosis when cells encounter outer membrane stresses. We tested whether morusin induced autophagy in A549 and NCI-H292 cells using immunoblot analysis and fluorescence microscopy. LC3-II and SQSTM1 are two widely used markers to reflect the progression of autophagy [25,26]. Firstly, protein levels of LC3-II and SQSTM1 in morusin treated cells were detected. As shown in Fig. 3A, after 0.5 h treatment with morusin, increased LC3-II and decreased SQSTM1 levels with a dose-dependent manner were observed in both cell lines. Next, a

valuable tool for monitoring autophagic flux, tandem RFP-GFP-LC3 construct was utilized. In normal condition, the LC3-II positive autophagosomes are labeled with yellow, after fusion with lysosomes, autolysosomes are shown as red puncta, indicate the induction of autophagic flux [27]. As observed under fluorescence microscopy (Fig. 3B), the increased red puncta in morusin treated group represented the formation of autolysosomes, confirming that autophagic flux was induced. An autophagic flux inhibitor chloroquine was used to evaluate

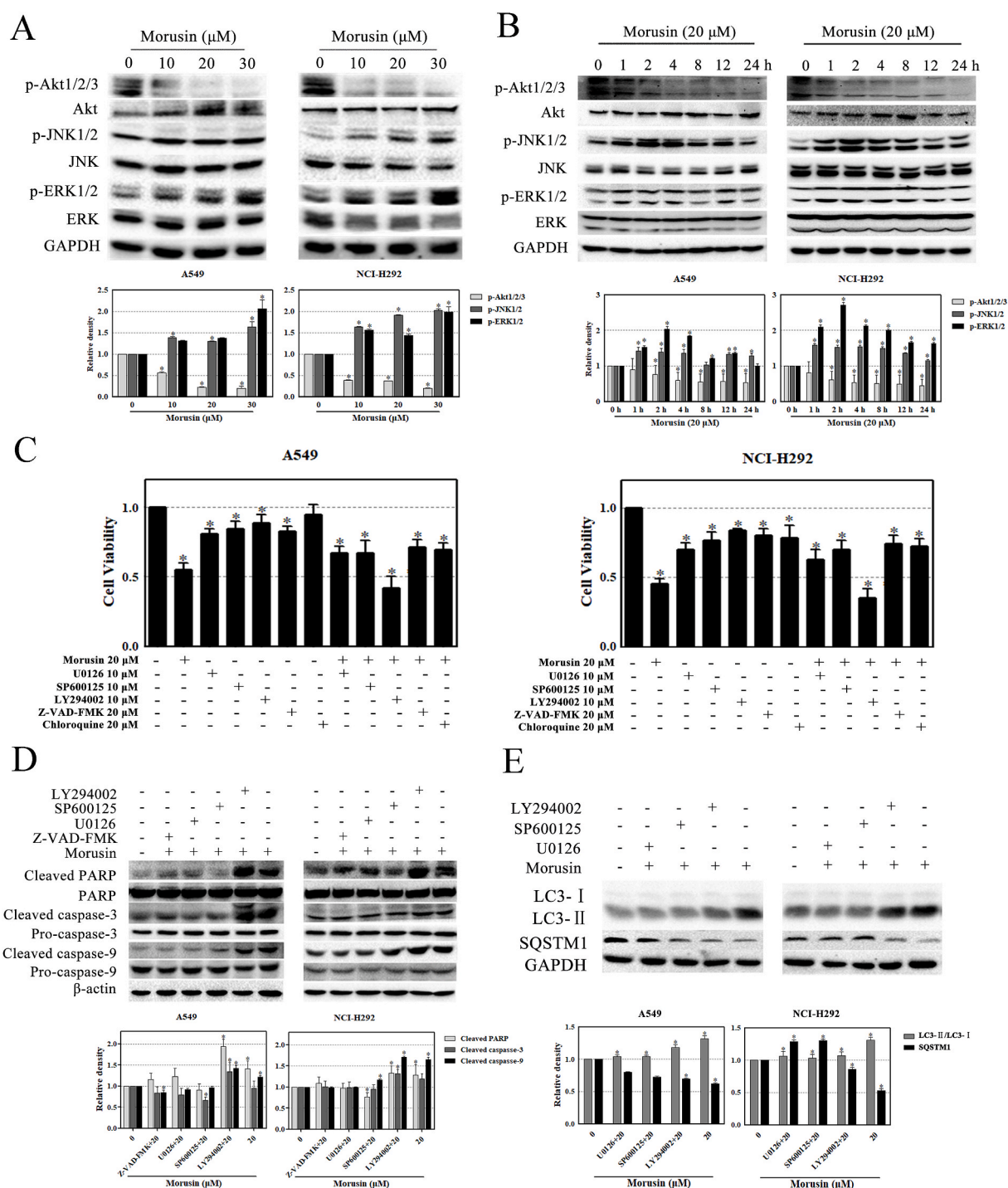


Fig. 4. PI3K-Akt suppression, ERK and JNK activation were involved in morusin-induced apoptosis and autophagy in A549 and NCI-H292 cells. (A) Cells were treated with morusin at different concentrations for 2 h and (B) were treated with morusin of 20 μM for different times, cell lysates were subjected to immunoblotting for detection of indicated proteins, and GAPDH was used as control. Bar graphs represented the relative express level of proteins. (C, D and E) Cells were pretreated with indicated inhibitors for 1 h, then treated with morusin for 48 h, cell activity was detected by CCK-8 assay (C); and the levels of apoptosis related (D) and autophagy related proteins (E) were detected. Values are expressed as mean $\pm$ SD, $n = 3$, $*p < 0.05$ versus control group.

the role of autophagy in morusin induced cell viability impairment, as shown in Fig. 4C, co-treatment of morusin with chloroquine rescued cell viability impairment induced by morusin partially, indicated that morusin induced autophagic flux contributed to the inhibition of cell growth in A549 and NCI-H292 cells.

3.5. Morusin suppressed Akt and activated JNK, ERK in A549 and NCI-H292 cells

Mitogen-activated protein kinase (MAPK) and phosphatidylinositol 3-hydroxy kinase (PI3K) signaling pathways are closely related to cell survival and death. In order to elucidate the mechanism of morusin-induced lung cancer cell death, we examined the expression levels of the representative proteins in these pathways [28,29]. As shown in

Fig. 4A and B, phosphorylation of Akt was inhibited dose- and time-dependently after morusin treatment, meanwhile phosphorylation of JNK or ERK were induced in dose- and time-dependent manner, indicated that morusin suppressed PI3K/Akt pathway, while activated JNK and ERK pathways. To elucidate the involvement of these pathways with morusin induced cell growth inhibition effects, inhibitors of JNK (SP600125), PI3K/Akt (LY294002) and MEK/ERK (U0126) were used. The combination of SP600125 or U0126 with morusin antagonized the cytotoxic effect of morusin, while combined treatment of LY194002 and morusin induced a synergistic effect (Fig. 4C). Based on these results, we further investigated the influence of these inhibitors on the expression of some proteins in apoptosis and autophagy pathways. As shown in Fig. 4D, combined treatment of morusin with U0126 or SP600125 reduced the expression of cleaved PARP, caspase-3 and caspase-9

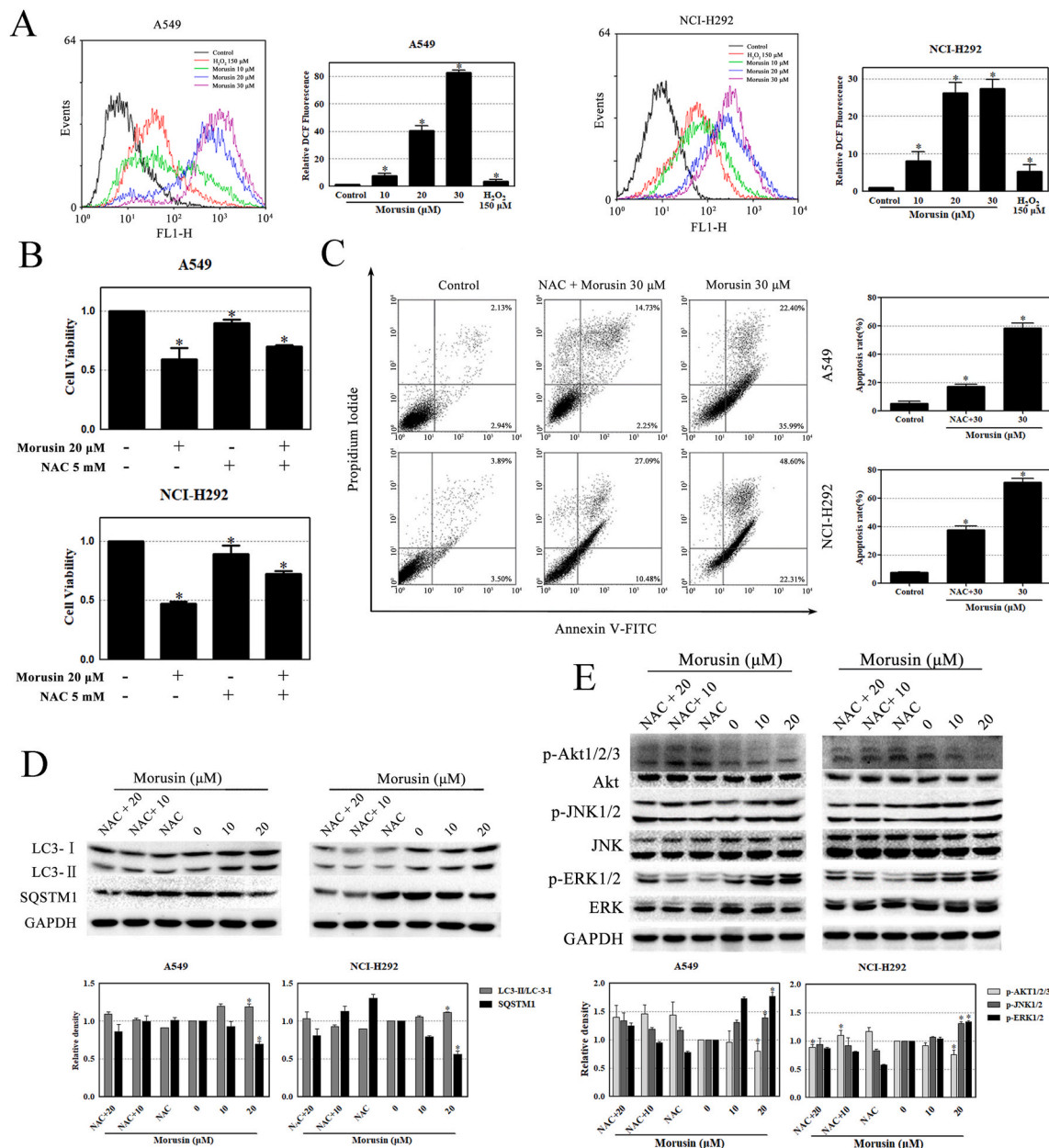


Fig. 5. ROS generation was involved in morusin induced apoptosis and autophagy. (A) A549 and NCI-H292 cells were treated with morusin for 24 h, the levels of ROS were determined by flow cytometry. H_2O_2 treatment was used as positive control. (B and C) Cells were pretreated with 5 mM NAC for 1 h, and then treated with 30 μ M morusin for 48 h. Subsequently, cell viability was detected by CCK-8 assay (B) and apoptosis rate of cells was analyzed by flow cytometry after Annexin V/PI staining (C). (D) The level of LC3 and SQSTM1 proteins and (E) phospho-ERK1/2, phospho-JNK1/2 and phospho-Akt1/2/3 proteins were detected post co-treatment of NAC (incubating in advance for 1 h) and morusin with 0.5 h and 2 h respectively. Values above are expressed as mean $\pm$ SD, $n = 3$, $*p < 0.05$.

compared to morusin treatment alone, while co-treatment of morusin with LY294002 did the opposite effect, that is, enhanced the expression of cleaved PARP, caspase-3 and caspase-9, these results were coincident with the results of cell viability assay. For the autophagy pathway, morusin treatment with U0126 or SP600125 decreased LC3-II expression, while increased SQSTM1 expression, indicated that blocking morusin-induced activation of JNK or ERK inhibited the occurrence of autophagy (Fig. 4E). For the co-treatment with LY294002, there was not obvious change of the expression of both LC3-II and SQSTM1 compared to morusin treatment alone, this maybe because morusin already inhibited the phosphorylation of Akt dramatically, LY294002 could not exert further inhibition (Fig. 4E). These results suggested that morusin induced suppression of Akt and activation of JNK and ERK pathways involved in both apoptosis and autophagy in A549 and NCI-H292 cells.

3.6. ROS generation is involved in morusin-induced apoptosis and autophagy in A549 and NCI-H292 cells

Accumulating evidences indicated that increased intracellular ROS level could induce cell death through various mechanisms, including apoptosis and autophagy [30]. We examined the effect of morusin on ROS generation in A549 and NCI-H292 cells. DCHF-DA was used as molecular probe for the flow cytometry detection. The results indicated that in the cells treated with different concentrations of morusin for 24 h, the ROS level increased dramatically in a dose-dependent manner (Fig. 5A). Combined treatment with a ROS inhibitor, N-Acetyl-L-Cysteine (NAC) inhibited morusin-induced cell death, suggested that ROS contributed to the cytotoxicity of morusin (Fig. 5B). Further investigation indicated that NAC inhibited morusin-induced cell apoptosis, as

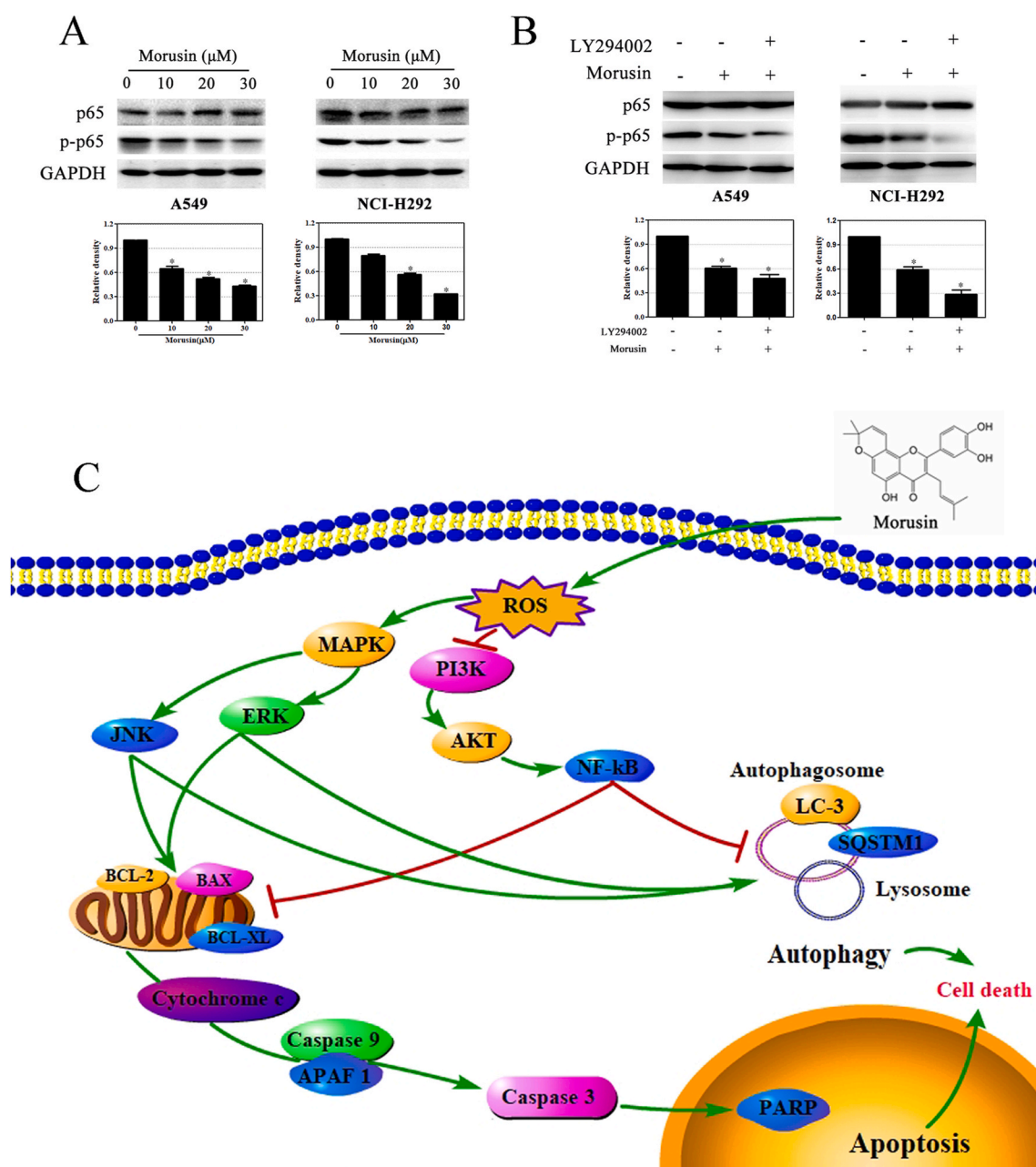


Fig. 6. Effects of morusin on phosphorylation of p65 and proposed mechanism of morusin on lung cancer cells. (A) A549 and NCI-H292 cells were treated with morusin for 2 h, and (B) were pretreated with LY294002 for 1 h, then treated with 20 μM morusin for additional 2 h. Cell lysates were subjected to western blot for p65 and phosphor-p65, and GAPDH was used as control. Bar graphs represented the relative express level of phosphor-p65. (C) Schematic depicting the effect of morusin on cell survival and underlying mechanism.

evidenced by the PI/Annexin V double staining result (Fig. 5C). NAC also attenuated morusin-induced LC3-II upregulation and SQSTM1 downregulation, suggested that ROS also contributed to morusin-induced autophagy (Fig. 5D).

Moreover, NAC was used to verify the role of ROS on Akt, JNK and ERK pathways. Co-treatment of A549 and NCI-H292 cells with NAC weakened the suppressed effect of morusin on the protein expression of p-Akt, also weakened the induced effects on protein level of p-JNK and p-ERK (Fig. 5E). This result suggested that morusin promoted generation of ROS probably involved in the suppression of Akt and activation of JNK and ERK.

3.7. Morusin decreased NF- κ B p65 phosphorylation in A549 and NCI-H292 cells

NF- κ B is known to be involved in regulating cell survival, proliferation and metastasis. Studies have shown that PI3K/Akt pathway could activate NF- κ B, which mean NF- κ B is a downstream target of Akt [31, 32]. When A549 and NCI-H292 cells were treated with morusin at dose of 10, 20, 30 μ M, the results showed that the expression levels of phosphor-p65 was decreased, although the total level of NF- κ B p65 did not change obviously (Fig. 6A). When cells were co-treated with morusin 20 μ M and LY294002, the expression of phosphor-p65 decreased more than morusin treated alone group (Fig. 6B). This NF- κ B inhibition effect was consistent with Akt inhibition.

4. Discussion

As the leading cause of cancer mortality worldwide, lung cancer contributed one-quarter of all cancer deaths. Although there has been a steady increase in survival for most cancer types since the mid-1970s, advances have been slow for lung cancer [33]. Recent years, targeted agents such as EGFR inhibitors showed promising effects in the treatment of advanced non-small-cell lung cancer, but this was mainly limited for cancers with EGFR mutation [34]. For most types of lung cancers, developing new therapeutics that can serve as alternative strategic options is still in an urgent need. Morusin is a prenylated flavonoid obtained from the root bark of *Morus alba*. According to previous reports, morusin possessed anti-cancer activities against several kinds of cancer cells [10,11,13]. In this study, we demonstrated that morusin activated both apoptosis and autophagy in two kinds of non-small-cell lung cancer cells, A549 and NCI-H292.

Apoptosis is a programmed cellular process that occurs under either physiological or pathological conditions. Induction of cancer cell apoptosis is the eventual aim of many cancer therapies. Annexin V/PI double staining and DAPI staining results indicated that morusin induced apoptosis in the two cell lines. The signaling pathways for apoptosis are divided into two distinct categories: intrinsic and extrinsic pathways [35,36]. The intrinsic pathway is also called the mitochondrial pathway due to the involvement of mitochondria. In this study, the cleavage of caspase-9, loss of MMP, released cytochrome c into cytosol, increased level of Bax and decreased level of Bcl-2 revealed that morusin induced apoptosis was through intrinsic mitochondrial apoptotic pathway.

Although apoptosis was induced by morusin treatment, we noticed that the apoptosis rate was lower than the cell death rate, this reminded us there might be other cell death pathway except for apoptosis. Autophagy is another mode of programmed cell death and usually acts before apoptosis occurred when cells encounter outer membrane stresses. LC3-II and SQSTM1 are representative markers of autophagy [26]. After morusin treatment for half an hour, the protein levels of LC3-II and SQSTM1 were detected to be regulated inversely, LC3-II increased and SQSTM1 decreased, this suggested the occurring of autophagy (Fig. 3A). To further verify the induction of autophagy, tandem RFP-GFP-LC3 was transfected into cells, immunofluorescence images indicated the induction of autophagic flux (Fig. 3B). Since autophagy can act both as a

cell survival and cell death process, we used an autophagy inhibitor chloroquine to demonstrate that morusin-induced autophagy promoted cell death (Fig. 4C).

ERK and JNK, known as members of MAPK family, play a key role in regulating cell growth, proliferation and cell survival [37]. PI3K-Akt pathway also plays an important role in controlling cell survival and death [38]. In this study, we examined the phosphorylation status of ERK, JNK and Akt to elucidate the possible mechanisms that are involved in morusin-induced apoptosis and autophagy. The results showed that morusin phosphorylated ERK and JNK, but dephosphorylated Akt (Fig. 4A and B). As an Akt downstream target, NF- κ B was shown to be inhibited accordingly (Fig. 6A). The use of corresponding inhibitors of each pathway revealed the involvement of ERK, JNK and Akt pathways in morusin-induced apoptosis and autophagy (Fig. 4D and E).

All of the pathways mentioned above are reported to be responsive to multiple stress stimuli including oxidative stress [39]. In this study, ROS generation was detected as increased significantly after morusin treatment. Cotreatment with NAC reversed the apoptosis and autophagy induced by morusin which indicated that ROS was an involved factor of apoptosis and autophagy induction (Fig. 5C and D).

In summary, the present study verified that morusin effectively decreased cell viability through inducing endogenous apoptosis and increasing autophagy flux in A549 and NCI-H292 cells. PI3K/Akt suppression and MAPK ERK, JNK activation contributed to apoptosis and autophagy inducing effects. The increased intracellular ROS generation involved in the regulation of PI3K/Akt and MAPK ERK, JNK, this might be the possible mechanism of action of morusin (Fig. 6C). This study provided support that morusin deserved further study for the development as a candidate for the treatment of lung cancer.

CRediT authorship contribution statement

Jinxia Wang: Data curation, Formal analysis, Writing - original draft, performed the experiments, collected and analyzed data, wrote the paper. **Xiaoqing Liu:** performed the experiments, collected and analyzed data. **Hao Zheng:** performed the experiments. **Huaran Zhang:** performed the experiments. **Xiaoning Wang:** performed the experiments, conceived and designed the research. **Tao Shen:** conceived and designed the research. **Shuqi Wang:** Data curation, Formal analysis, conceived and designed the research, collected and analyzed data. **Dongmei Ren:** Data curation, Formal analysis, Writing - original draft, conceived and designed the research, collected and analyzed data, wrote the paper..

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cbi.2020.109279>.

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